

Lecture # 24

Function generator module.

Date: 11-11-25

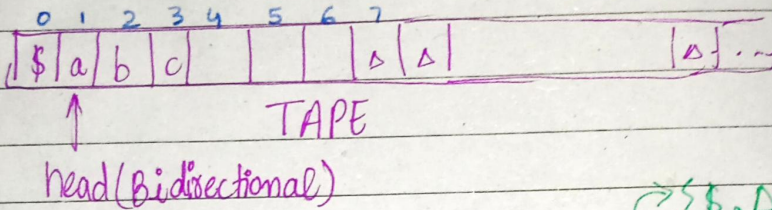
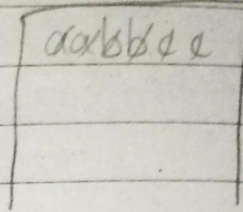
Turing Machine

Day: Tuesday

↳ FA + Tape

$$d = \{a^n b^m c^n : n \geq 0\}$$

- read/modify
- Movement
- state of mind



4 E 7

$\Phi = \Phi_1 \Phi_2 \Phi_3$

$$\Gamma = \Gamma_{sc} U \Sigma$$
 $\rightarrow \{b, \Delta\}$

1. (tape elem)

directions:

Transition function:

$$Q \rtimes \Gamma \longrightarrow Q' \rtimes \Gamma \rtimes D$$

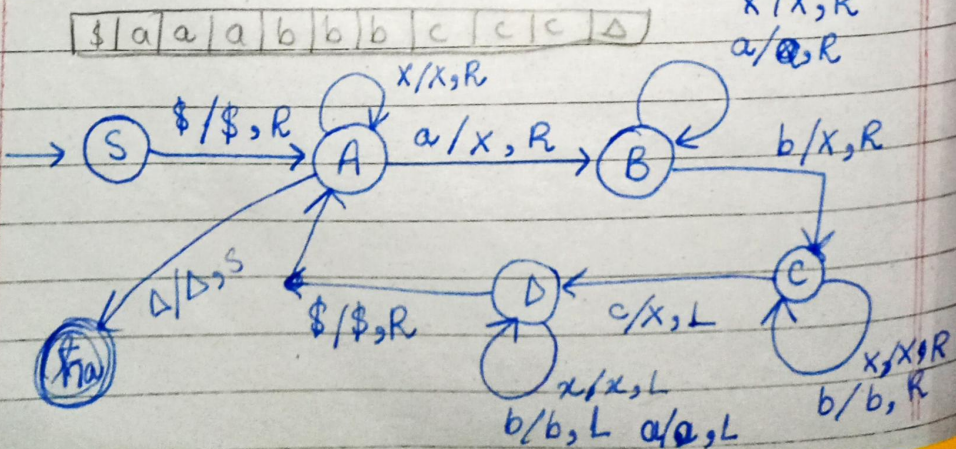
→ read

→ direction

 $\rightarrow \{L, R, S\}$

→ haunting
state
for
rejection

haunting
state for
acceptance

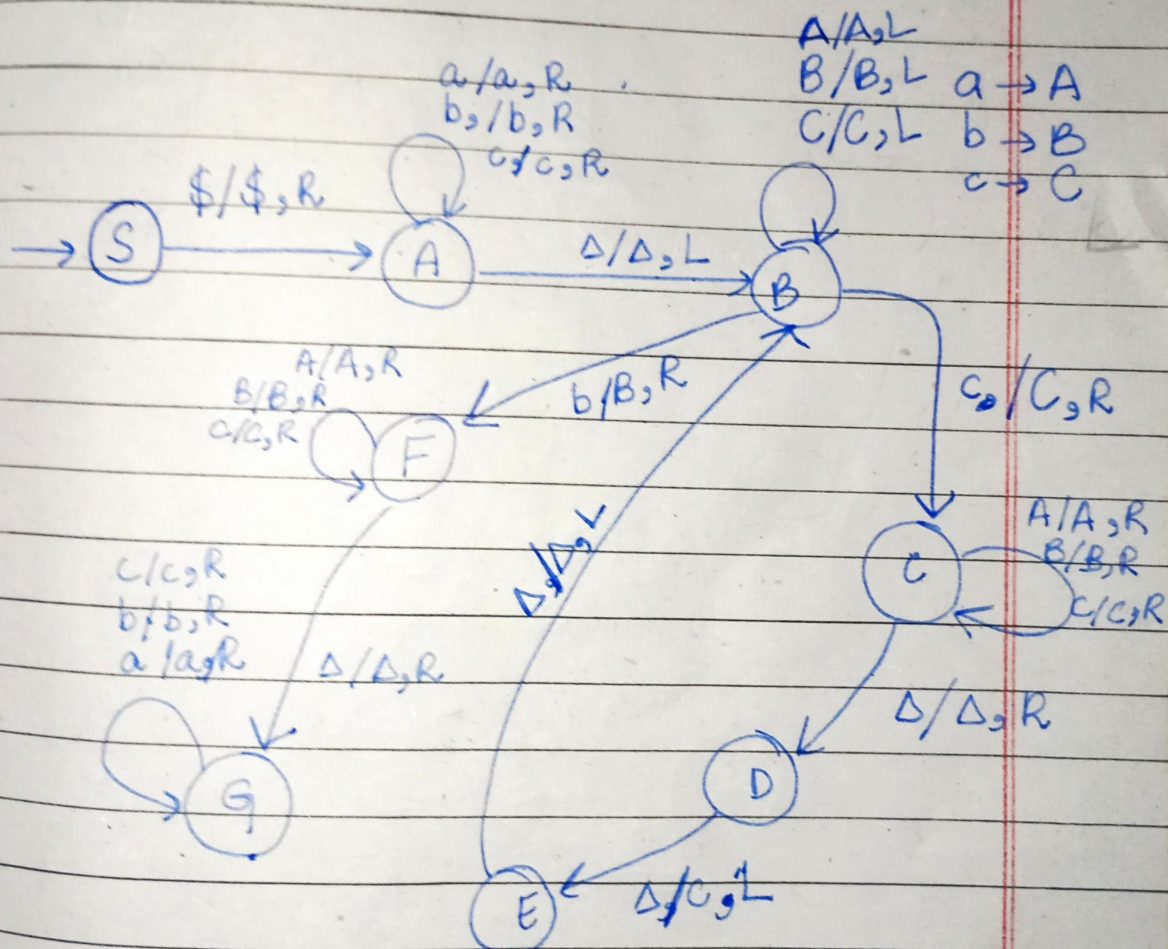
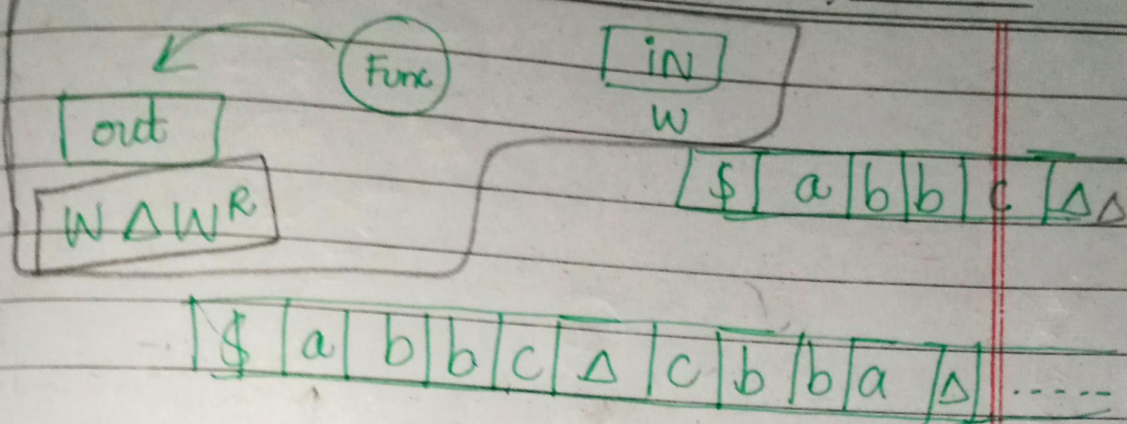
$$X/X, R$$
 $a/\mathbb{Q}, R$ 

X NOR,

2,

Date:

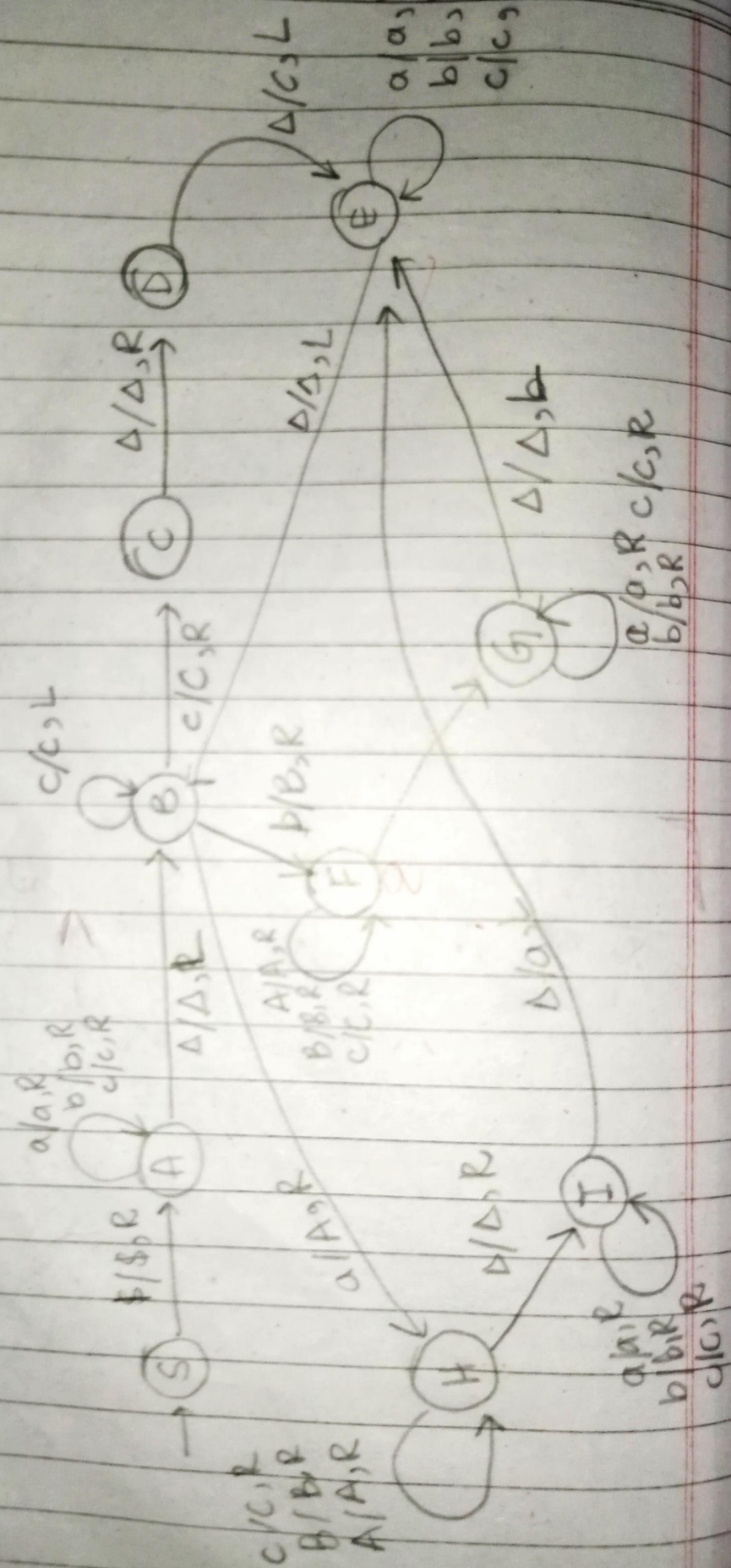
Day:



WΔWR

a	b	b	c	c	Δ	c
					↑	↑

Day: _____

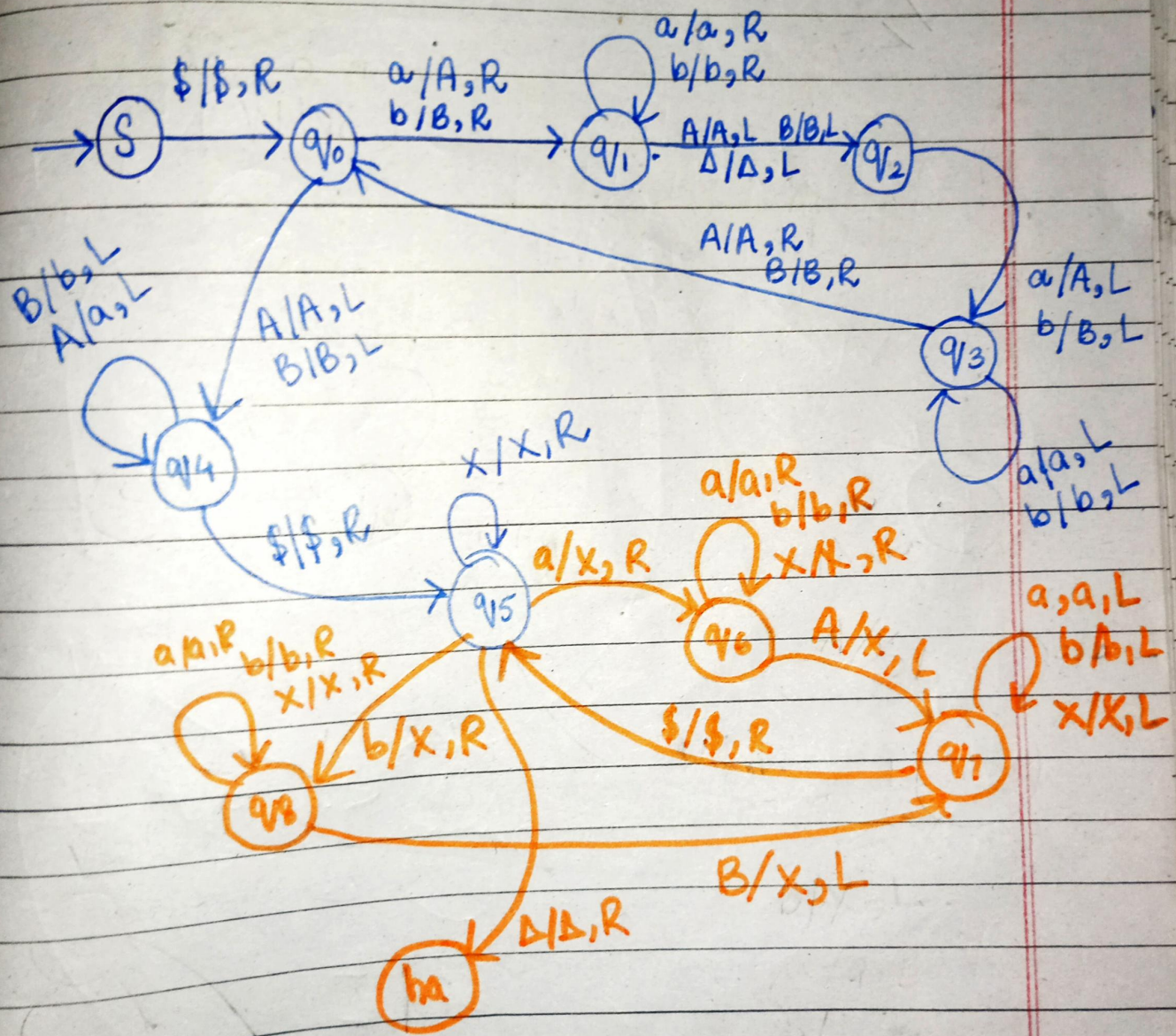


b/b
a/a

$$L = \{ ww \mid w \text{ over } \{a, b\} \}$$

\$ | a | b | b | a | b | b | Δ | ... |

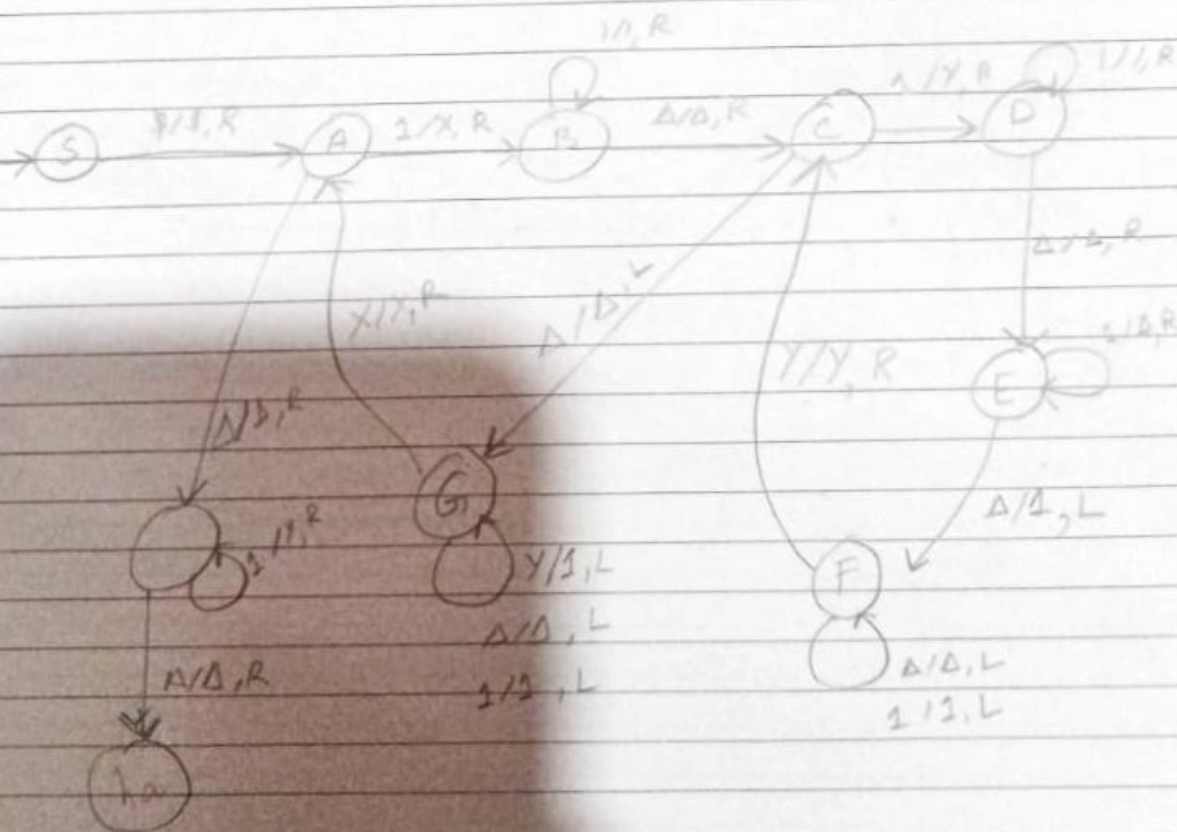
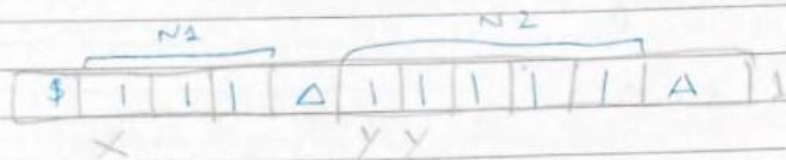
check if it's valid.



$$X \Delta Y = \text{Result}$$

Date: _____

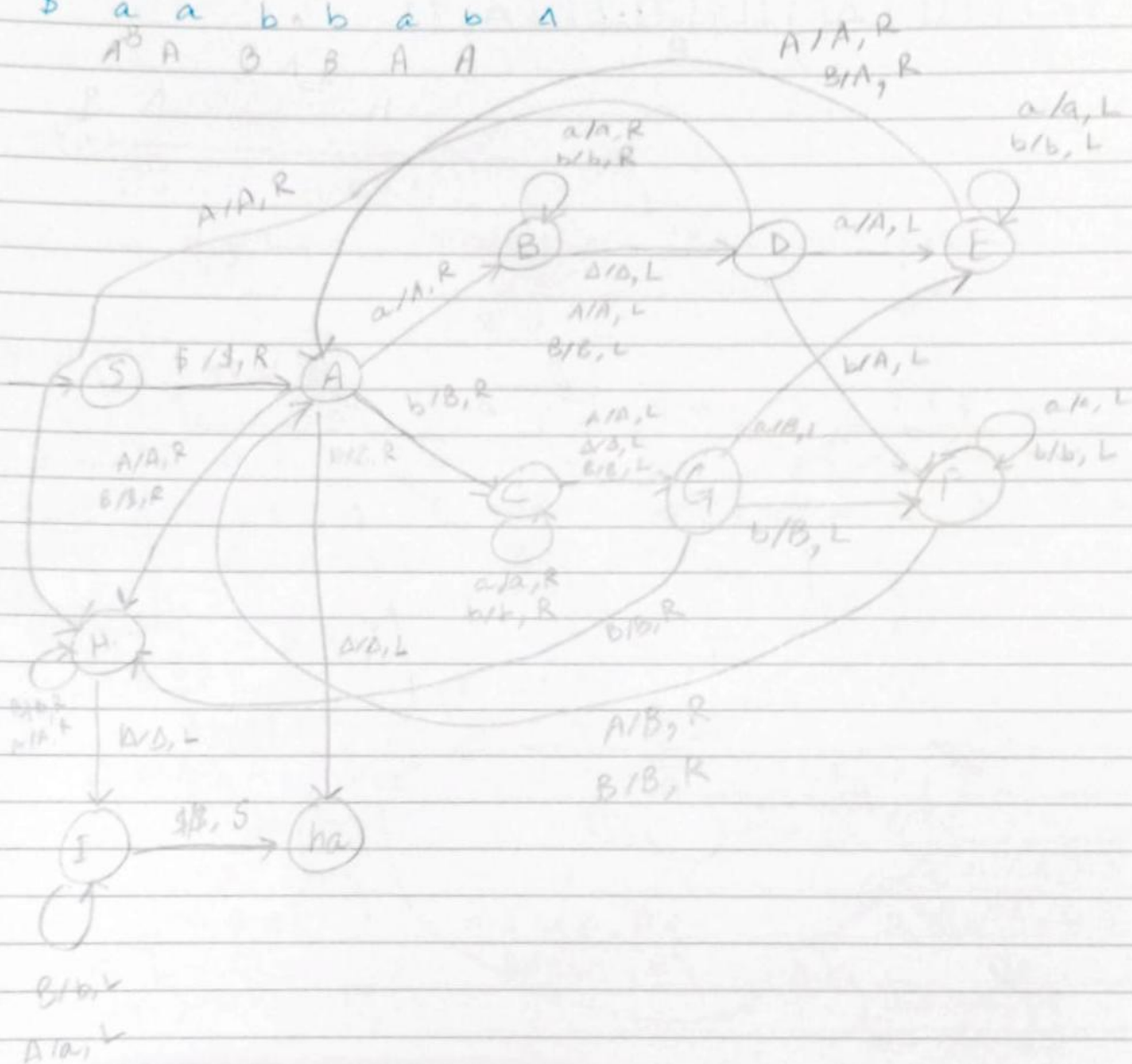
TM for Multiplication



Date: 18/11/25

TM to reverse a string

\$ a a b b a b a ...
 $\overline{A}^0 A \quad \overline{B} \quad \overline{B} \quad A \quad A$



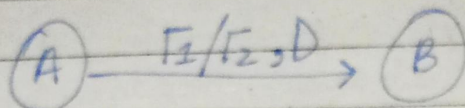
Lecture # 27

Date: Nov 20th, 2025

Day: Thursday

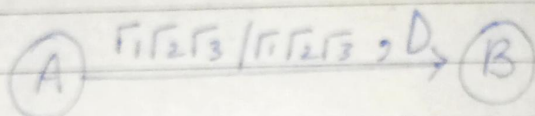
Multitap Turing Machine (multiple taps with multiple pointers)

in STTM:



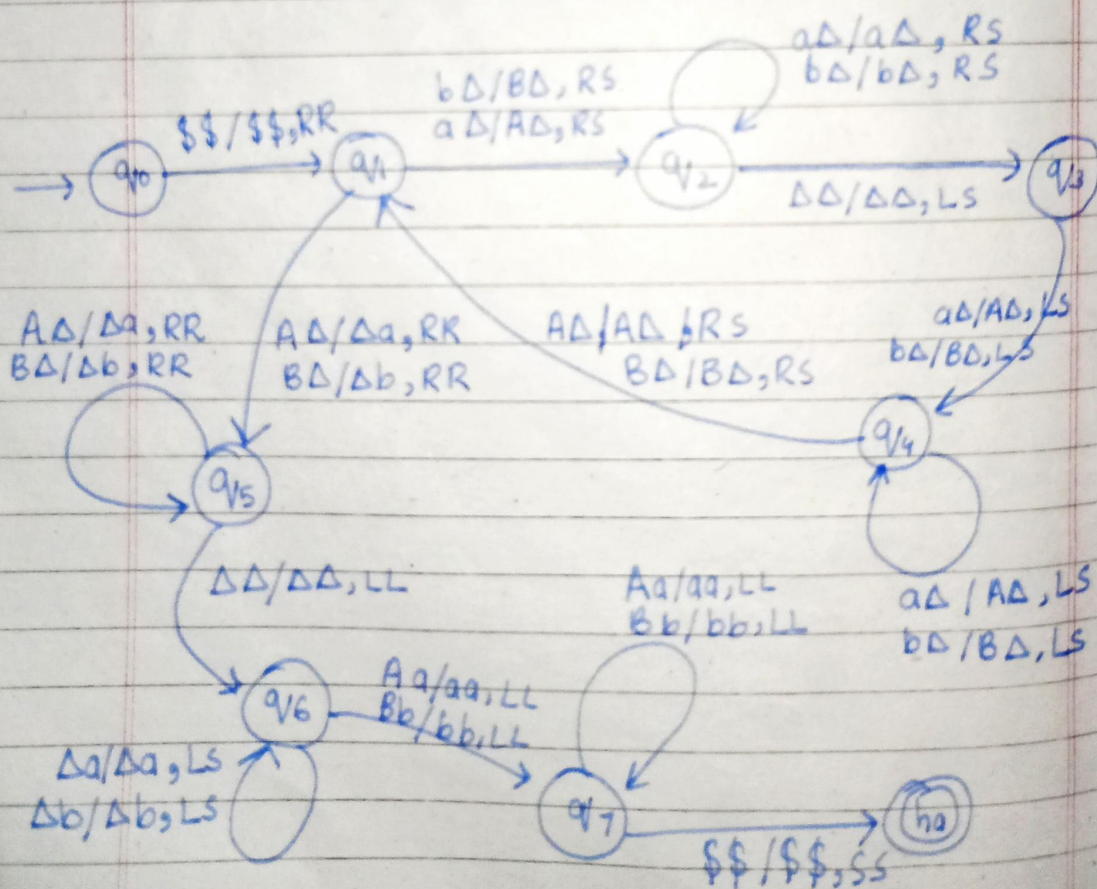
in MTTM: (A) $\xrightarrow{(T_1/T_2/T_3)/(T_1/T_2/T_3), D_1/D_2/D_3} (B)$

Multitrack:



WW:

T_1 :	\$	a	b	b	a	b	b	Δ
T_2 :	\$							



Date: _____

Day: _____

n1:

\$	Δ	Δ	Δ	1	1	Δ
\$	1	1	1	Δ	Δ	Δ
\$						

n2:

n3 : n1 • n2:

